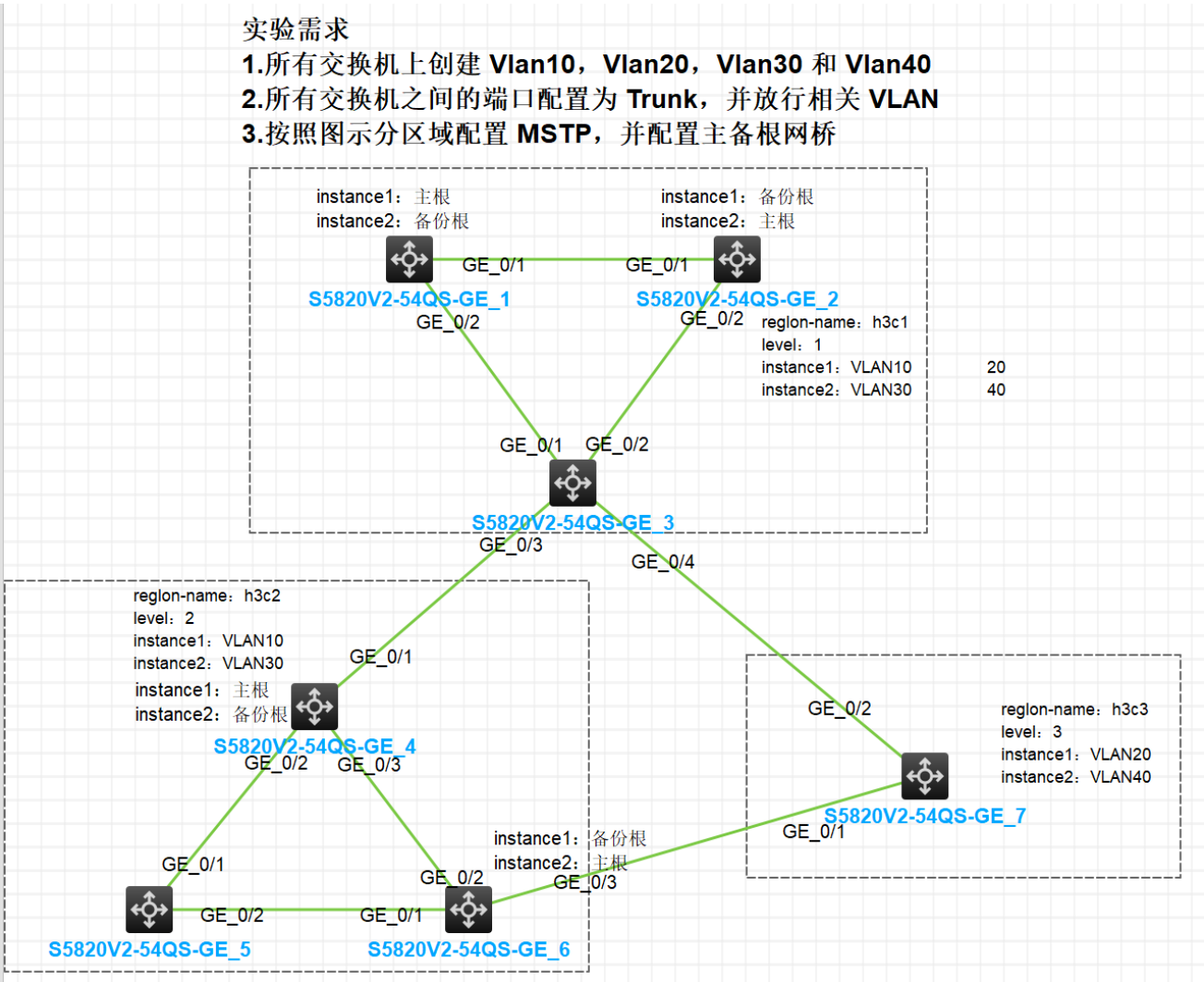


MSTP实验

实验拓扑



实验需求

1. 所有交换机上创建 Vlan10, Vlan20, Vlan30 和 Vlan40
2. 所有交换机之间的端口配置为 Trunk, 并放行相关 VLAN
3. 按照图示分区域配置 MSTP, 并配置主备根网桥

实验配置

```
[SW1]vlan 10
[SW1-vlan10]vlan 20
[SW1-vlan20]vlan 30
[SW1-vlan30]vlan 40

[SW1-vlan40]int ra g 1/0/1 to g 1/0/2
```

```

[SW1-if-range]port link-type trunk
[SW1-if-range]port trunk permit vlan 10 20 30 40
[SW1]stp region-configuration
[SW1-mst-region]region-name h3c1
[SW1-mst-region]revision-level 1
[SW1-mst-region]instance 1 vlan 10 20
[SW1-mst-region]instance 2 vlan 30 40
[SW1-mst-region]active region-configuration
[SW1]stp instance 1 root primary
[SW1]stp instance 2 root secondary
---
[SW2]vlan 10
[SW2-vlan10]vlan 20
[SW2-vlan20]vlan 30
[SW2-vlan30]vlan 40
[SW2-vlan40]int ra g 1/0/1 to g 1/0/2
[SW2-if-range]port link-type trunk
[SW2-if-range]port trunk permit vlan 10 20 30 40
[SW2]stp region-configuration
[SW2-mst-region]region-name h3c1
[SW2-mst-region]revision-level 1
[SW2-mst-region]instance 1 vlan 10 20
[SW2-mst-region]instance 2 vlan 30 40
[SW2-mst-region]active region-configuration
[SW2]stp instance 1 root secondary
[SW2]stp instance 2 root primary
---
[SW3]vlan 10
[SW3-vlan10]vlan 20
[SW3-vlan20]vlan 30
[SW3-vlan30]vlan 40
[SW3-vlan40]int ra g 1/0/1 to g 1/0/2
[SW3-if-range]port link-type trunk
[SW3-if-range]port trunk permit vlan 10 20 30 40
[SW3-if-range]int g1/0/3
[SW3-GigabitEthernet1/0/3]port link-type trunk
[SW3-GigabitEthernet1/0/3]port trunk permit vlan 10 30
[SW3-GigabitEthernet1/0/3]int g1/0/4
[SW3-GigabitEthernet1/0/3]port link-type trunk
[SW3-GigabitEthernet1/0/3]port trunk permit vlan 20 40
[SW3]stp region-configuration
[SW3-mst-region]region-name h3c1
[SW3-mst-region]revision-level 1
[SW3-mst-region]instance 1 vlan 10 20
[SW3-mst-region]instance 2 vlan 30 40
[SW3-mst-region]active region-configuration
---
[SW4]vlan 10
[SW4-vlan10]vlan 20
[SW4-vlan20]vlan 30
[SW4-vlan30]vlan 40
[SW4-vlan40]int ra g 1/0/2 to g 1/0/3

[SW4-if-range]port link-type trunk

```

```

[SW4-if-range]port trunk permit vlan 10 30
[SW4-if-range]int g1/0/1
[SW4-GigabitEthernet1/0/1]port link-type trunk
[SW4-GigabitEthernet1/0/1]port trunk permit vlan 10 30
[SW4]stp region-configuration
[SW4-mst-region]region-name h3c2
[SW4-mst-region]revision-level 2
[SW4-mst-region]instance 1 vlan 10
[SW4-mst-region]instance 2 vlan 30
[SW4-mst-region]active region-configuration
[SW4]stp instance 1 root primary
[SW4]stp instance 2 root secondary
---
[SW5]vlan 10
[SW5-vlan10]vlan 20
[SW5-vlan20]vlan 30
[SW5-vlan30]vlan 40
[SW5-vlan40]int ra g 1/0/1 to g 1/0/2
[SW5-if-range]port link-type trunk
[SW5-if-range]port trunk permit vlan 10 30
[SW5]stp region-configuration
[SW5-mst-region]region-name h3c2
[SW5-mst-region]revision-level 2
[SW5-mst-region]instance 1 vlan 10
[SW5-mst-region]instance 2 vlan 30
[SW5-mst-region]active region-configuration
---
[SW6]vlan 10
[SW6-vlan10]vlan 20
[SW6-vlan20]vlan 30
[SW6-vlan30]vlan 40
[SW6-vlan40]int ra g 1/0/1 to g 1/0/2
[SW6-if-range]port link-type trunk
[SW6-if-range]port trunk permit vlan 10 30
[SW6-if-range]int g1/0/3
[SW6-GigabitEthernet1/0/3]port link-type trunk
[SW6-GigabitEthernet1/0/3]port trunk permit vlan 10 30
[SW6]stp region-configuration
[SW6-mst-region]region-name h3c2
[SW6-mst-region]revision-level 2
[SW6-mst-region]instance 1 vlan 10
[SW6-mst-region]instance 2 vlan 30
[SW6-mst-region]active region-configuration
[SW6]stp instance 1 root secondary
[SW6]stp instance 2 root primary
---
[SW7]vlan 10
[SW7-vlan10]vlan 20
[SW7-vlan20]vlan 30
[SW7-vlan30]vlan 40
[SW7-vlan40]int ra g 1/0/1 to g 1/0/2
[SW7-if-range]port link-type trunk

[SW7-if-range]port trunk permit vlan 20 40

```

```
[SW7]stp region-configuration
[SW7-mst-region]region-name h3c3
[SW7-mst-region]revision-level 3
[SW7-mst-region]instance 1 vlan 20
[SW7-mst-region]instance 2 vlan 40
[SW7-mst-region]active region-configuration
```

结果查看

1.查看端口角色

```
<SW1>display stp brief
```

MST ID	Port	Role	STP State	Protection
0	GigabitEthernet1/0/1	DESI	FORWARDING	NONE
0	GigabitEthernet1/0/2	DESI	FORWARDING	NONE
1	GigabitEthernet1/0/1	DESI	FORWARDING	NONE
1	GigabitEthernet1/0/2	DESI	FORWARDING	NONE
2	GigabitEthernet1/0/1	ROOT	FORWARDING	NONE
2	GigabitEthernet1/0/2	DESI	FORWARDING	NONE

```
---
```

```
<SW2>display stp brief
```

MST ID	Port	Role	STP State	Protection
0	GigabitEthernet1/0/1	ROOT	FORWARDING	NONE
0	GigabitEthernet1/0/2	DESI	FORWARDING	NONE
1	GigabitEthernet1/0/1	ROOT	FORWARDING	NONE
1	GigabitEthernet1/0/2	DESI	FORWARDING	NONE
2	GigabitEthernet1/0/1	DESI	FORWARDING	NONE
2	GigabitEthernet1/0/2	DESI	FORWARDING	NONE

```
---
```

```
<SW3>display stp brief
```

MST ID	Port	Role	STP State	Protection
0	GigabitEthernet1/0/1	ROOT	FORWARDING	NONE
0	GigabitEthernet1/0/2	ALTE	DISCARDING	NONE
0	GigabitEthernet1/0/3	DESI	FORWARDING	NONE
0	GigabitEthernet1/0/4	DESI	FORWARDING	NONE
1	GigabitEthernet1/0/1	ROOT	FORWARDING	NONE
1	GigabitEthernet1/0/2	ALTE	DISCARDING	NONE
1	GigabitEthernet1/0/3	DESI	FORWARDING	NONE
2	GigabitEthernet1/0/1	ALTE	DISCARDING	NONE
2	GigabitEthernet1/0/2	ROOT	FORWARDING	NONE
2	GigabitEthernet1/0/3	DESI	FORWARDING	NONE

```
---
```

```
<SW4>display stp brief
```

MST ID	Port	Role	STP State	Protection
0	GigabitEthernet1/0/1	ROOT	FORWARDING	NONE
0	GigabitEthernet1/0/2	DESI	FORWARDING	NONE
0	GigabitEthernet1/0/3	DESI	FORWARDING	NONE
1	GigabitEthernet1/0/1	MAST	FORWARDING	NONE
1	GigabitEthernet1/0/2	DESI	FORWARDING	NONE

```

1      GigabitEthernet1/0/3      DESI FORWARDING NONE
2      GigabitEthernet1/0/1      MAST FORWARDING NONE
2      GigabitEthernet1/0/2      DESI FORWARDING NONE
2      GigabitEthernet1/0/3      ROOT FORWARDING NONE
---
<SW5>display stp brief
MST ID  Port                      Role  STP State  Protection
0      GigabitEthernet1/0/1      ROOT  FORWARDING NONE
0      GigabitEthernet1/0/2      DESI  FORWARDING NONE
1      GigabitEthernet1/0/1      ROOT  FORWARDING NONE
1      GigabitEthernet1/0/2      ALTE  DISCARDING NONE
2      GigabitEthernet1/0/1      ALTE  DISCARDING NONE
2      GigabitEthernet1/0/2      ROOT  FORWARDING NONE
---
<SW6>display stp brief
MST ID  Port                      Role  STP State  Protection
0      GigabitEthernet1/0/1      ALTE  DISCARDING NONE
0      GigabitEthernet1/0/2      ROOT  FORWARDING NONE
0      GigabitEthernet1/0/3      DESI  FORWARDING NONE
1      GigabitEthernet1/0/1      DESI  FORWARDING NONE
1      GigabitEthernet1/0/2      ROOT  FORWARDING NONE
1      GigabitEthernet1/0/3      DESI  FORWARDING NONE
2      GigabitEthernet1/0/1      DESI  FORWARDING NONE
2      GigabitEthernet1/0/2      DESI  FORWARDING NONE
2      GigabitEthernet1/0/3      DESI  FORWARDING NONE
---
<SW7>display stp brief
MST ID  Port                      Role  STP State  Protection
0      GigabitEthernet1/0/1      ALTE  DISCARDING NONE
0      GigabitEthernet1/0/2      ROOT  FORWARDING NONE
1      GigabitEthernet1/0/1      ALTE  DISCARDING NONE
1      GigabitEthernet1/0/2      MAST  FORWARDING NONE
2      GigabitEthernet1/0/1      ALTE  DISCARDING NONE
2      GigabitEthernet1/0/2      MAST  FORWARDING NONE

```

2.查看root和实例

```

<SW1>display stp root
MST ID  Root Bridge ID      ExtPathCost  IntPathCost  Root Port
0      32768.78e6-afe2-0100  0            0
1      0.78e6-afe2-0100    0            0
2      0.78e6-b1a7-0200    0            20          GE1/0/1
<SW1>display stp region-configuration
Oper Configuration
Format selector      : 0
Region name          : h3c1
Revision level       : 1
Configuration digest : 0xca136a235706b316c8db8f921067a68f

Instance  VLANs Mapped
0         1 to 9, 11 to 19, 21 to 29, 31 to 39, 41 to 4094
1         10, 20

```

```

2          30, 40
---
<SW2>display stp root
MST ID    Root Bridge ID          ExtPathCost IntPathCost Root Port
0          32768.78e6-afe2-0100  0            20           GE1/0/1
1          0.78e6-afe2-0100      0            20           GE1/0/1
2          0.78e6-b1a7-0200      0            0
<SW2>display stp region-configuration
Oper Configuration
  Format selector      : 0
  Region name         : h3c1
  Revision level       : 1
  Configuration digest : 0xca136a235706b316c8db8f921067a68f

Instance VLANs Mapped
0          1 to 9, 11 to 19, 21 to 29, 31 to 39, 41 to 4094
1          10, 20
2          30, 40
---
<SW3>display stp root
MST ID    Root Bridge ID          ExtPathCost IntPathCost Root Port
0          32768.78e6-afe2-0100  0            20           GE1/0/1
1          0.78e6-afe2-0100      0            20           GE1/0/1
2          0.78e6-b1a7-0200      0            20           GE1/0/2
<SW3>display stp region-configuration
Oper Configuration
  Format selector      : 0
  Region name         : h3c1
  Revision level       : 1
  Configuration digest : 0xca136a235706b316c8db8f921067a68f

Instance VLANs Mapped
0          1 to 9, 11 to 19, 21 to 29, 31 to 39, 41 to 4094
1          10, 20
2          30, 40
---
<SW4>display stp root
MST ID    Root Bridge ID          ExtPathCost IntPathCost Root Port
0          32768.78e6-afe2-0100  20           0            GE1/0/1
1          0.78e6-b56e-0400      0            0            GE1/0/1
2          0.78e6-b8c5-0600      0            20           GE1/0/3
<SW4>display stp region-configuration
Oper Configuration
  Format selector      : 0
  Region name         : h3c2
  Revision level       : 2
  Configuration digest : 0x18591c17578097f2cb778f123b16efae

Instance VLANs Mapped
0          1 to 9, 11 to 29, 31 to 4094
1          10
2          30
---

```

```

<SW5>display stp root
MST ID    Root Bridge ID          ExtPathCost IntPathCost Root Port
0         32768.78e6-afe2-0100 20           20           GE1/0/1
1         0.78e6-b56e-0400    0           20           GE1/0/1
2         0.78e6-b8c5-0600    0           20           GE1/0/2
<SW5>display stp region-configuration
Oper Configuration
Format selector      : 0
Region name          : h3c2
Revision level       : 2
Configuration digest : 0x18591c17578097f2cb778f123b16efae

Instance VLANs Mapped
0         1 to 9, 11 to 29, 31 to 4094
1         10
2         30
---
```

```

<SW6>display stp root
MST ID    Root Bridge ID          ExtPathCost IntPathCost Root Port
0         32768.78e6-afe2-0100 20           20           GE1/0/2
1         0.78e6-b56e-0400    0           20           GE1/0/2
2         0.78e6-b8c5-0600    0           0
<SW6>display stp region-configuration
Oper Configuration
Format selector      : 0
Region name          : h3c2
Revision level       : 2
Configuration digest : 0x18591c17578097f2cb778f123b16efae

Instance VLANs Mapped
0         1 to 9, 11 to 29, 31 to 4094
1         10
2         30
---
```

```

<SW7>display stp root
MST ID    Root Bridge ID          ExtPathCost IntPathCost Root Port
0         32768.78e6-afe2-0100 20           0           GE1/0/2
1         32768.78e6-ba6a-0700 0           0
2         32768.78e6-ba6a-0700 0           0
<SW7>display stp region-configuration
Oper Configuration
Format selector      : 0
Region name          : h3c3
Revision level       : 3
Configuration digest : 0xdd5b4ddb5bdf444012bfcc12f9d36d8c

Instance VLANs Mapped
0         1 to 19, 21 to 39, 41 to 4094
1         20
2         40

```

3.查看实例1和2

```

<SW1>display stp instance 1
-----[MSTI 1 Global Info]-----
Bridge ID           : 0.78e6-afe2-0100
RegRoot ID/IRPC     : 0.78e6-afe2-0100, 0
RootPort ID        : 0.0
Root type           : Primary root
Master bridge       : 32768.78e6-afe2-0100
Cost to master      : 0
TC received         : 0

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----
Port protocol       : Enabled
Port role           : Designated Port
Port ID             : 128.2
Port cost(Legacy)    : Config=auto, Active=20
Desg.bridge/port    : 0.78e6-afe2-0100, 128.2
Protection type     : Config=none, Active=none
Rapid transition    : False
Num of VLANs mapped : 2
Port times          : RemHops 20

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
Port protocol       : Enabled
Port role           : Designated Port
Port ID             : 128.3
Port cost(Legacy)    : Config=auto, Active=20
Desg.bridge/port    : 0.78e6-afe2-0100, 128.3
Protection type     : Config=none, Active=none
Rapid transition    : False
Num of VLANs mapped : 2
Port times          : RemHops 20
<SW1>display stp instance 2
-----[MSTI 2 Global Info]-----
Bridge ID           : 4096.78e6-afe2-0100
RegRoot ID/IRPC     : 0.78e6-b1a7-0200, 20
RootPort ID        : 128.2
Root type           : Secondary root
Master bridge       : 32768.78e6-afe2-0100
Cost to master      : 0
TC received         : 0

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----
Port protocol       : Enabled
Port role           : Root Port
Port ID             : 128.2
Port cost(Legacy)    : Config=auto, Active=20
Desg.bridge/port    : 0.78e6-b1a7-0200, 128.2
Protection type     : Config=none, Active=none
Rapid transition    : False
Num of VLANs mapped : 2
Port times          : RemHops 20

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----

```



```
Port protocol      : Enabled
Port role          : Designated Port
Port ID            : 128.3
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 4096.78e6-afe2-0100, 128.3
Protection type    : Config=none, Active=none
Rapid transition   : False
Num of VLANs mapped : 2
Port times         : RemHops 19
```

```
<SW2>display stp instance 1
```

```
-----[MSTI 1 Global Info]-----
```

```
Bridge ID          : 4096.78e6-b1a7-0200
RegRoot ID/IRPC    : 0.78e6-afe2-0100, 20
RootPort ID        : 128.2
Root type          : Secondary root
Master bridge      : 32768.78e6-afe2-0100
Cost to master     : 20
TC received        : 0
```

```
----[Port2(GigabitEthernet1/0/1)][FORWARDING]----
```

```
Port protocol      : Enabled
Port role          : Root Port
Port ID            : 128.2
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 0.78e6-afe2-0100, 128.2
Protection type    : Config=none, Active=none
Rapid transition   : True
Num of VLANs mapped : 2
Port times         : RemHops 20
```

```
----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
```

```
Port protocol      : Enabled
Port role          : Designated Port
Port ID            : 128.3
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 4096.78e6-b1a7-0200, 128.3
Protection type    : Config=none, Active=none
Rapid transition   : False
Num of VLANs mapped : 2
Port times         : RemHops 19
```

```
<SW2>display stp instance 2
```

```
-----[MSTI 2 Global Info]-----
```

```
Bridge ID          : 0.78e6-b1a7-0200
RegRoot ID/IRPC    : 0.78e6-b1a7-0200, 0
RootPort ID        : 0.0
Root type          : Primary root
Master bridge      : 32768.78e6-afe2-0100
Cost to master     : 20
TC received        : 0
```

```
----[Port2(GigabitEthernet1/0/1)][FORWARDING]----
```

```
Port protocol      : Enabled
```

```
Port role          : Designated Port
Port ID            : 128.2
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 0.78e6-b1a7-0200, 128.2
Protection type    : Config=none, Active=none
Rapid transition   : True
Num of VLANs mapped : 2
Port times         : RemHops 20
```

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----

```
Port protocol      : Enabled
Port role          : Designated Port
Port ID            : 128.3
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 0.78e6-b1a7-0200, 128.3
Protection type    : Config=none, Active=none
Rapid transition   : False
Num of VLANs mapped : 2
Port times         : RemHops 20
```

```
<SW3>display stp instance 1
```

-----[MSTI 1 Global Info]-----

```
Bridge ID          : 32768.78e6-b31d-0300
RegRoot ID/IRPC    : 0.78e6-afe2-0100, 20
RootPort ID        : 128.2
Master bridge      : 32768.78e6-afe2-0100
Cost to master      : 20
TC received         : 0
```

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----

```
Port protocol      : Enabled
Port role          : Root Port
Port ID            : 128.2
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 0.78e6-afe2-0100, 128.3
Protection type    : Config=none, Active=none
Rapid transition   : True
Num of VLANs mapped : 2
Port times         : RemHops 20
```

----[Port3(GigabitEthernet1/0/2)][DISCARDING]----

```
Port protocol      : Enabled
Port role          : Alternate Port
Port ID            : 128.3
Port cost(Legacy)  : Config=auto, Active=20
Desg.bridge/port   : 4096.78e6-b1a7-0200, 128.3
Protection type    : Config=none, Active=none
Rapid transition   : False
Num of VLANs mapped : 2
Port times         : RemHops 19
```

----[Port4(GigabitEthernet1/0/3)][FORWARDING]----

```
Port protocol      : Enabled
```

```

Port role           : Designated Port (Boundary)
Port ID             : 128.4
Port cost(Legacy)   : Config=auto, Active=20
Desg.bridge/port    : 32768.78e6-b31d-0300, 128.4
Protection type     : Config=none, Active=none
Rapid transition    : False
Num of VLANs mapped : 2
Port times          : RemHops 19
<SW3>display stp instance 2
-----[MSTI 2 Global Info]-----
Bridge ID           : 32768.78e6-b31d-0300
RegRoot ID/IRPC     : 0.78e6-b1a7-0200, 20
RootPort ID         : 128.3
Master bridge       : 32768.78e6-afe2-0100
Cost to master      : 20
TC received         : 0

----[Port2(GigabitEthernet1/0/1)][DISCARDING]----
Port protocol       : Enabled
Port role           : Alternate Port
Port ID             : 128.2
Port cost(Legacy)   : Config=auto, Active=20
Desg.bridge/port    : 4096.78e6-afe2-0100, 128.3
Protection type     : Config=none, Active=none
Rapid transition    : True
Num of VLANs mapped : 2
Port times          : RemHops 19

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
Port protocol       : Enabled
Port role           : Root Port
Port ID             : 128.3
Port cost(Legacy)   : Config=auto, Active=20
Desg.bridge/port    : 0.78e6-b1a7-0200, 128.3
Protection type     : Config=none, Active=none
Rapid transition    : True
Num of VLANs mapped : 2
Port times          : RemHops 20

----[Port4(GigabitEthernet1/0/3)][FORWARDING]----
Port protocol       : Enabled
Port role           : Designated Port (Boundary)
Port ID             : 128.4
Port cost(Legacy)   : Config=auto, Active=20
Desg.bridge/port    : 32768.78e6-b31d-0300, 128.4
Protection type     : Config=none, Active=none
Rapid transition    : False
Num of VLANs mapped : 2
Port times          : RemHops 19
---
<SW4>display stp instance 1
-----[MSTI 1 Global Info]-----
Bridge ID           : 0.78e6-b56e-0400

```

```
RegRoot ID/IRPC      : 0.78e6-b56e-0400, 0
RootPort ID          : 0.0
Root type             : Primary root
Master bridge         : 32768.78e6-b56e-0400
Cost to master        : 0
TC received           : 0
```

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----

```
Port protocol        : Enabled
Port role             : Master Port (Boundary)
Port ID               : 128.2
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port      : 0.78e6-b56e-0400, 128.2
Protection type       : Config=none, Active=none
Rapid transition      : True
Num of VLANs mapped  : 1
Port times            : RemHops 20
```

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----

```
Port protocol        : Enabled
Port role             : Designated Port
Port ID               : 128.3
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port      : 0.78e6-b56e-0400, 128.3
Protection type       : Config=none, Active=none
Rapid transition      : True
Num of VLANs mapped  : 1
Port times            : RemHops 20
```

----[Port4(GigabitEthernet1/0/3)][FORWARDING]----

```
Port protocol        : Enabled
Port role             : Designated Port
Port ID               : 128.4
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port      : 0.78e6-b56e-0400, 128.4
Protection type       : Config=none, Active=none
Rapid transition      : False
Num of VLANs mapped  : 1
Port times            : RemHops 20
```

<SW4>display stp instance 2

-----[MSTI 2 Global Info]-----

```
Bridge ID             : 4096.78e6-b56e-0400
RegRoot ID/IRPC       : 0.78e6-b8c5-0600, 20
RootPort ID           : 128.4
Root type              : Secondary root
Master bridge          : 32768.78e6-b56e-0400
Cost to master         : 0
TC received            : 0
```

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----

```
Port protocol        : Enabled
Port role             : Master Port (Boundary)

Port ID               : 128.2
```

```
Port cost(Legacy)      : Config=auto, Active=20
Desg.bridge/port      : 4096.78e6-b56e-0400, 128.2
Protection type       : Config=none, Active=none
Rapid transition      : True
Num of VLANs mapped  : 1
Port times            : RemHops 19
```

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----

```
Port protocol         : Enabled
Port role             : Designated Port
Port ID              : 128.3
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port      : 4096.78e6-b56e-0400, 128.3
Protection type       : Config=none, Active=none
Rapid transition      : True
Num of VLANs mapped  : 1
Port times            : RemHops 19
```

----[Port4(GigabitEthernet1/0/3)][FORWARDING]----

```
Port protocol         : Enabled
Port role             : Root Port
Port ID              : 128.4
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port      : 0.78e6-b8c5-0600, 128.3
Protection type       : Config=none, Active=none
Rapid transition      : False
Num of VLANs mapped  : 1
Port times            : RemHops 20
```

<SW5>display stp instance 1

-----[MSTI 1 Global Info]-----

```
Bridge ID            : 32768.78e6-b72b-0500
RegRoot ID/IRPC      : 0.78e6-b56e-0400, 20
RootPort ID          : 128.2
Master bridge        : 32768.78e6-b56e-0400
Cost to master       : 20
TC received          : 0
```

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----

```
Port protocol         : Enabled
Port role             : Root Port
Port ID              : 128.2
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port      : 0.78e6-b56e-0400, 128.3
Protection type       : Config=none, Active=none
Rapid transition      : True
Num of VLANs mapped  : 1
Port times            : RemHops 20
```

----[Port3(GigabitEthernet1/0/2)][DISCARDING]----

```
Port protocol         : Enabled
Port role             : Alternate Port

Port ID              : 128.3
```

```

Port cost(Legacy)      : Config=auto, Active=20
Desg.bridge/port      : 4096.78e6-b8c5-0600, 128.2
Protection type       : Config=none, Active=none
Rapid transition      : False
Num of VLANs mapped  : 1
Port times            : RemHops 19
<SW5>display stp instance 2
-----[MSTI 2 Global Info]-----
Bridge ID             : 32768.78e6-b72b-0500
RegRoot ID/IRPC       : 0.78e6-b8c5-0600, 20
RootPort ID           : 128.3
Master bridge         : 32768.78e6-b56e-0400
Cost to master        : 20
TC received           : 0

----[Port2(GigabitEthernet1/0/1)][DISCARDING]----
Port protocol        : Enabled
Port role            : Alternate Port
Port ID              : 128.2
Port cost(Legacy)    : Config=auto, Active=20
Desg.bridge/port     : 4096.78e6-b56e-0400, 128.3
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped  : 1
Port times           : RemHops 19

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
Port protocol        : Enabled
Port role            : Root Port
Port ID              : 128.3
Port cost(Legacy)    : Config=auto, Active=20
Desg.bridge/port     : 0.78e6-b8c5-0600, 128.2
Protection type      : Config=none, Active=none
Rapid transition     : False
Num of VLANs mapped  : 1
Port times           : RemHops 20
---
<SW6>display stp instance 1
-----[MSTI 1 Global Info]-----
Bridge ID             : 4096.78e6-b8c5-0600
RegRoot ID/IRPC       : 0.78e6-b56e-0400, 20
RootPort ID           : 128.3
Root type             : Secondary root
Master bridge         : 32768.78e6-b56e-0400
Cost to master        : 20
TC received           : 0

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----
Port protocol        : Enabled
Port role            : Designated Port
Port ID              : 128.2
Port cost(Legacy)    : Config=auto, Active=20
Desg.bridge/port     : 4096.78e6-b8c5-0600, 128.2

```

```
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 19

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
Port protocol        : Enabled
Port role            : Root Port
Port ID              : 128.3
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 0.78e6-b56e-0400, 128.4
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 20

----[Port4(GigabitEthernet1/0/3)][FORWARDING]----
Port protocol        : Enabled
Port role            : Designated Port (Boundary)
Port ID              : 128.4
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 4096.78e6-b8c5-0600, 128.4
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 19
<SW6>display stp instance 2
-----[MSTI 2 Global Info]-----
Bridge ID            : 0.78e6-b8c5-0600
RegRoot ID/IRPC      : 0.78e6-b8c5-0600, 0
RootPort ID          : 0.0
Root type            : Primary root
Master bridge        : 32768.78e6-b56e-0400
Cost to master       : 20
TC received          : 0

----[Port2(GigabitEthernet1/0/1)][FORWARDING]----
Port protocol        : Enabled
Port role            : Designated Port
Port ID              : 128.2
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 0.78e6-b8c5-0600, 128.2
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 20

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
Port protocol        : Enabled
Port role            : Designated Port
Port ID              : 128.3
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 0.78e6-b8c5-0600, 128.3
```

```

Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 20

----[Port4(GigabitEthernet1/0/3)][FORWARDING]----
Port protocol        : Enabled
Port role            : Designated Port (Boundary)
Port ID              : 128.4
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 0.78e6-b8c5-0600, 128.4
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 20
---
<SW7>display stp instance 1
-----[MSTI 1 Global Info]-----
Bridge ID            : 32768.78e6-ba6a-0700
RegRoot ID/IRPC      : 32768.78e6-ba6a-0700, 0
RootPort ID          : 0.0
Master bridge        : 32768.78e6-ba6a-0700
Cost to master       : 0
TC received          : 0

----[Port2(GigabitEthernet1/0/1)][DISCARDING]----
Port protocol        : Enabled
Port role            : Alternate Port (Boundary)
Port ID              : 128.2
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 32768.78e6-ba6a-0700, 128.2
Protection type      : Config=none, Active=none
Rapid transition     : False
Num of VLANs mapped : 1
Port times           : RemHops 20

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----
Port protocol        : Enabled
Port role            : Master Port (Boundary)
Port ID              : 128.3
Port cost(Legacy)     : Config=auto, Active=20
Desg.bridge/port     : 32768.78e6-ba6a-0700, 128.3
Protection type      : Config=none, Active=none
Rapid transition     : True
Num of VLANs mapped : 1
Port times           : RemHops 20
<SW7>display stp instance 2
-----[MSTI 2 Global Info]-----
Bridge ID            : 32768.78e6-ba6a-0700
RegRoot ID/IRPC      : 32768.78e6-ba6a-0700, 0
RootPort ID          : 0.0
Master bridge        : 32768.78e6-ba6a-0700
Cost to master       : 0

```


TC received : 0

----[Port2(GigabitEthernet1/0/1)][DISCARDING]----

Port protocol : Enabled
Port role : Alternate Port (Boundary)
Port ID : 128.2
Port cost(Legacy) : Config=auto, Active=20
Desg.bridge/port : 32768.78e6-ba6a-0700, 128.2
Protection type : Config=none, Active=none
Rapid transition : False
Num of VLANs mapped : 1
Port times : RemHops 20

----[Port3(GigabitEthernet1/0/2)][FORWARDING]----

Port protocol : Enabled
Port role : Master Port (Boundary)
Port ID : 128.3
Port cost(Legacy) : Config=auto, Active=20
Desg.bridge/port : 32768.78e6-ba6a-0700, 128.3
Protection type : Config=none, Active=none
Rapid transition : True
Num of VLANs mapped : 1
Port times : RemHops 20